

CE 503: Fundamentals of Project Management

Course Code: CE 503

Course Name: Fundamentals of Project Management

L-T-P-C: 3-0-0-3

Prerequisite: None

Students intended for: UG/PG

Elective or Core: Elective

Approval: 9th Senate

Course content

- **Introduction**

Project definition, Constraints and Scope triangle, Project environment, Classification of projects, Conception, Selection, Life cycle, Project management – necessity and processes, Systems approach, Project manager – knowledge areas and role, Project team, Planning – principles, objectives, steps and advantages, Work breakdown structure, Scheduling - Bar charts, Milestone charts, Networks. Project control and evaluation. [8 Lectures]

- **Scheduling by network analysis**

(i) Programme Evaluation and Review Technique (PERT): Time estimates for activities, Computation of event times, Network analysis – Slack and critical path. (ii) Critical Path Method (CPM): Computation of event and activity times, Network analysis – Float and critical path. [6 Lectures]

- **Time-Cost optimization**

Crashing a network. Updating a project: Data required and implementation. Resource allocation: Resource usage profiles, Smoothing and Levelling of resources. [4 Lectures]

- **Quality management**

Policy, Assurance, Management systems, Control, Plan, Audit, Reviews, Statistical methods for quality control – Shewhart, Cusum and EWMA control charts, Process capability analysis, Factorial experiments and Acceptance sampling. [10 Lectures]

- **Management science techniques**

Linear and integer programming, Distribution and network models – transportation, assignment, transshipment and shortest route problems. Non-linear optimization, Time series analysis and forecasting. [14 Lectures]

Text Books

1. B.C.Punmia and K.K.Khandelwal, **Project planning and control with PERT and CPM**, Laxmi Publications, 2006.
2. E. L. Grant and R. S. Leavenworth, **Statistical quality control**, 7th Edition, TMH, 2000.
3. D. R. Anderson, D. J. Sweeney, T. A. Williams, J. D. Camm and Kipp Martin, **An introduction to management science**, 13th Edition, South-Western, 2012.

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4. Hamdy A. Taha, **Operations Research – an introduction**, Pearson. 2014.
 5. J. M. Nicholas and Herman Steyn, **Project management for engineering, business and technology**, 4th Edition, Routledge, 2012.

Reference Books:

1. James P. Lewis, **Fundamentals of project management**, 3rd Edition, AMACOM, 2007.
2. Albert Lester, **Project management, planning and control**, Elsevier, 2014.
3. D. C. Montgomery, **Introduction to statistical quality control**, 6th Edition, Wiley, 2009.
4. J. A. Lawrence, Jr. and B. A. Pasternack, **Applied management science**, Wiley, 2002.
5. J. D. West and F. K. Levy, **A management guide to PERT/CPM with GERT/PDM/DCPM**, 1979.